

Set Theory, Relations and Inclusion–Exclusion

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What is a Set?

Definition

A **set** is a well-defined collection of distinct objects.

The objects in a set are called **elements** or **members** of the set.

For example,

$$A = \{1, 2, 3, 4, 5\}.$$

Here,

$$1, 2, 3, 4, 5 \in A.$$

If x does not belong to A , we write

$$x \notin A.$$

Set-Builder Form

Definition

Set-builder form describes a set by stating the property or condition satisfied by its elements.

It is generally written as

$$A = \{x \mid P(x)\}.$$

Example:

$$A = \{x \mid x \in \mathbb{N}, 1 \leq x \leq 5\}.$$

This means that A contains all natural numbers from 1 to 5.

Therefore,

$$A = \{1, 2, 3, 4, 5\}.$$

Definition

Roster form represents a set by listing all its elements inside braces.

Example:

$$A = \{1, 2, 3, 4, 5\}.$$

The same set can be written in set-builder form as

$$A = \{x \mid x \in \mathbb{N}, 1 \leq x \leq 5\}.$$

Thus, set-builder form gives a **condition**, while roster form gives the **elements explicitly**.

Set-Builder and Roster Form: Examples 1–5

No.	Set-Builder Form	Roster Form
1	$\{x \mid x \in \mathbb{N}, 1 \leq x \leq 5\}$	$\{1, 2, 3, 4, 5\}$
2	$\{x \mid x \text{ is an even number}, 2 \leq x \leq 10\}$	$\{2, 4, 6, 8, 10\}$
3	$\{x \mid x \text{ is an odd number}, 1 \leq x \leq 9\}$	$\{1, 3, 5, 7, 9\}$
4	$\{x \mid x \text{ is a prime number}, x < 15\}$	$\{2, 3, 5, 7, 11, 13\}$
5	$\{x \mid x = n^2, n \in \{1, 2, 3, 4\}\}$	$\{1, 4, 9, 16\}$

Set-Builder and Roster Form: Examples 6–10

No.	Set-Builder Form	Roster Form
6	$\{x \mid x \text{ is a positive multiple of } 3, x \leq 15\}$	$\{3, 6, 9, 12, 15\}$
7	$\{x \mid x \text{ is a vowel in the English alphabet}\}$	$\{a, e, i, o, u\}$
8	$\{x \mid x \text{ is a day starting with 's'}\}$	$\{\text{Saturday, Sunday}\}$
9	$\{x \mid x \text{ is a month having 30 days}\}$	$\{\text{April, June, September, November}\}$
10	$\{x \mid x \text{ is a lowercase English letter from } a \text{ to } d\}$	$\{a, b, c, d\}$

Types of Sets

Important types of sets are:

- 1 Cardinality
- 2 Infinite Set
- 3 Null Set
- 4 Equal Sets
- 5 Equivalent Sets
- 6 Subset
- 7 Superset
- 8 Universal Set
- 9 Power Set
- 10 Singleton Set
- 11 Countable Set

Cardinality and Infinite Set

Cardinality

The cardinality of a set is the number of elements in the set.

Let

$$A = \{\text{Linear Regression, Decision Tree, SVM, KNN}\}.$$

Then

$$|A| = 4.$$

Infinite Set

An infinite set contains infinitely many elements.

$$B = \{x \mid x \text{ is a possible real-valued neural network weight}\}.$$

Null Set and Equal Sets

Null Set

A set containing no elements is called a null or empty set.

$$C = \{x \mid x \text{ is both supervised and unsupervised simultaneously}\} = \emptyset.$$

Equal Sets

Two sets are equal if they contain exactly the same elements.

$$A = \{\text{CNN, RNN, Transformer}\}$$

$$B = \{\text{Transformer, CNN, RNN}\}.$$

Hence,

$$A = B.$$

Equivalent Sets

Definition

Two sets are equivalent if they have the same number of elements.

Let

$$A = \{\text{CNN}, \text{RNN}, \text{SVM}\}$$

and

$$B = \{\text{Python}, \text{Java}, \text{C++}\}.$$

Since

$$|A| = |B| = 3,$$

the sets are equivalent.

Definition

A set A is a subset of B if every element of A is also an element of B .

$$A \subseteq B.$$

Let

$$A = \{\text{CNN}, \text{RNN}\}$$

and

$$B = \{\text{CNN}, \text{RNN}, \text{Transformer}\}.$$

Therefore,

$$A \subseteq B.$$

Definition

If every element of A belongs to B , then B is a superset of A .

$$B \supseteq A.$$

Let

$$A = \{\text{CNN}, \text{RNN}\}$$

and

$$B = \{\text{CNN}, \text{RNN}, \text{Transformer}\}.$$

Therefore,

$$B \supseteq A.$$

Definition

A universal set is the set containing all objects under consideration.

For AI/ML,

$$U = \{\text{All AI/ML algorithms studied}\}.$$

The universal set depends on the context of the problem.

For example, if we are considering machine learning algorithms studied in a course, then U contains all those algorithms.

Definition

The power set of A , denoted by $P(A)$, is the set of all subsets of A .

If

$$A = \{\text{CNN}, \text{RNN}\},$$

then

$$P(A) = \{\emptyset, \{\text{CNN}\}, \{\text{RNN}\}, \{\text{CNN}, \text{RNN}\}\}.$$

For a set containing n elements,

$$|P(A)| = 2^n.$$

Singleton and Countable Sets

Singleton Set

A set containing exactly one element is called a singleton set.

$$\{\text{Transformer}\}.$$

Countable Set

A set whose elements can be listed or matched with natural numbers is called countable.

Set of programming languages used in AI

$$= \{\text{Python, R, Java, Julia}\}.$$

This is finite, hence countable.

AI/ML Examples of Types of Sets

Type	AI/ML Example
Cardinality	Set of ML algorithms has 4 elements
Infinite Set	Possible real-valued neural network weights
Null Set	Objects that are both supervised and unsupervised simultaneously
Equal Sets	{CNN,RNN,Transformer} in different orders
Equivalent Sets	3 ML algorithms and 3 programming languages
Subset	$\{\text{CNN,RNN}\} \subseteq \{\text{CNN,RNN,Transformer}\}$
Superset	$\{\text{CNN,RNN,Transformer}\} \supseteq \{\text{CNN,RNN}\}$
Universal Set	All AI/ML algorithms studied
Power Set	All subsets of {CNN,RNN}
Singleton Set	{Transformer}
Countable Set	{Python,R,Java,Julia}

Set Operations

Let

$$A = \{\text{CNN}, \text{RNN}, \text{Transformer}\}$$

be the set of deep learning models studied in an AI course, and

$$B = \{\text{Transformer}, \text{SVM}, \text{Decision Tree}\}$$

be the set of models used in a machine learning project.

The important set operations are:

- ① Union
- ② Intersection
- ③ Set Difference
- ④ Symmetric Difference
- ⑤ Complement
- ⑥ Cartesian Product

Definition

The union of two sets contains all unique elements from both sets.

$$A \cup B = \{\text{CNN, RNN, Transformer, SVM, Decision Tree}\}.$$

Interpretation:

All AI/ML models considered in either the course or the project.

Definition

The intersection of two sets contains the elements common to both sets.

$$A \cap B = \{\text{Transformer}\}.$$

Interpretation:

Transformer is common to both the course and the project.

Definition

The difference $A - B$ contains elements that belong to A but not to B .

Models present in the course but not in the project:

$$A - B = \{\text{CNN}, \text{RNN}\}.$$

Models present in the project but not in the course:

$$B - A = \{\text{SVM}, \text{Decision Tree}\}.$$

Symmetric Difference

Definition

The symmetric difference contains elements belonging to exactly one of the two sets.

It is denoted by

$$A \triangle B.$$

For the given AI/ML sets,

$$A \triangle B = \{\text{CNN, RNN, SVM, Decision Tree}\}.$$

Complement

Definition

The complement of A contains the elements of the universal set that are not in A .

Let

$$U = \{\text{CNN, RNN, Transformer, SVM, Decision Tree, KNN, Naive Bayes}\}.$$

Then

$$A' = U - A$$

$$= \{\text{SVM, Decision Tree, KNN, Naive Bayes}\}.$$

Interpretation: AI/ML models in the syllabus that are not deep learning models.

Cartesian Product

Definition

The Cartesian product consists of all ordered pairs (a, b) where $a \in A$ and $b \in B$.

$$A \times B = \left\{ \begin{array}{l} (\text{CNN}, \text{Transformer}), \\ (\text{CNN}, \text{SVM}), \\ (\text{CNN}, \text{Decision Tree}), \\ (\text{RNN}, \text{Transformer}), \\ (\text{RNN}, \text{SVM}), \\ (\text{RNN}, \text{Decision Tree}), \\ (\text{Transformer}, \text{Transformer}), \\ (\text{Transformer}, \text{SVM}), \\ (\text{Transformer}, \text{Decision Tree}) \end{array} \right\}.$$

There are $3 \times 3 = 9$ ordered pairs.

Definition

A relation R from a set A to a set B is a subset of the Cartesian product $A \times B$.

$$R \subseteq A \times B.$$

Example:

Let

$$A = \{1, 2, 3\}, \quad B = \{2, 4, 6\}.$$

Define the relation

$$aRb \quad \text{if} \quad b = 2a.$$

Then

$$R = \{(1, 2), (2, 4), (3, 6)\}.$$

Relations on a Set

Definition

A relation R on a set A is a subset of $A \times A$.

$$R \subseteq A \times A.$$

For example, let

$$A = \{1, 2, 3\}.$$

A relation on A can be

$$R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3)\}.$$

Relations are used to describe connections between elements of a set.

Properties of Relations

A relation on a set can have the following properties.

Reflexive

For every $a \in A$,

$$(a, a) \in R.$$

Symmetric

For all $a, b \in A$,

$$(a, b) \in R \Rightarrow (b, a) \in R.$$

Antisymmetric

For all $a, b \in A$,

$$(a, b), (b, a) \in R \Rightarrow a = b.$$

Transitive

For all $a, b, c \in A$

Equivalence Relation

Definition

A relation R on A is an **equivalence relation** if it is reflexive, symmetric and transitive.

$$\text{Equivalence Relation} = \text{Reflexive} + \text{Symmetric} + \text{Transitive}$$

Example:

Let

$$A = \{1, 2, 3\}$$

and define

$$aRb \quad \text{if} \quad a = b.$$

Then

$$R = \{(1, 1), (2, 2), (3, 3)\}.$$

This relation is reflexive, symmetric and transitive.

Hence, R is an equivalence relation.

Equivalence Relation: AI/ML Example

Consider a set of machine learning models:

$$A = \{\text{CNN}_1, \text{CNN}_2, \text{RNN}_1, \text{RNN}_2\}.$$

Define a relation R by

$$xRy$$

if x and y belong to the same model family.

Then,

$$\text{CNN}_1 R \text{CNN}_2$$

and

$$\text{RNN}_1 R \text{RNN}_2.$$

Models belonging to the same family form equivalence classes.

Thus, equivalence relations can be used to group objects according to a common property.

Equivalence Classes

Definition

For an equivalence relation R on A , the equivalence class of a is the set of all elements related to a .

It is denoted by

$$[a] = \{x \in A \mid xRa\}.$$

For example, under congruence modulo 3,

$$a \equiv b \pmod{3}$$

if

$$3 \mid (a - b).$$

The equivalence classes are

Partial Order

Definition

A relation R on a set A is a **partial order** if it is reflexive, antisymmetric and transitive.

Partial Order = Reflexive + Antisymmetric + Transitive

Example:

Let

$$A = \{1, 2, 3, 4, 5\}$$

with the relation $\leq$.

The relation $\leq$ is:

- Reflexive
- Antisymmetric
- Transitive

Therefore,

Definition

A **partially ordered set**, or **poset**, is a set together with a partial order relation.

It is represented as

$$(A, \preceq).$$

Examples:

$$(A, \leq)$$

where

$$A = \{1, 2, 3, 4, 5\}.$$

Another important example is

$$(P(A), \subseteq).$$

Hasse Diagram

Definition

A **Hasse diagram** is a graphical representation of a finite partially ordered set.

In a Hasse diagram:

- Smaller elements are placed below larger elements.
- Reflexive relations are omitted.
- Transitive relations are omitted.
- Only covering relations are shown.

For example,

$$1 < 2 < 3 < 4 < 5$$

can be represented using a Hasse diagram.

Hasse Diagram of $(A, \leq)$

Let

$$A = \{1, 2, 3, 4, 5\}.$$

The relation is $\leq$.

The covering relations are

$$1 < 2, \quad 2 < 3, \quad 3 < 4, \quad 4 < 5.$$



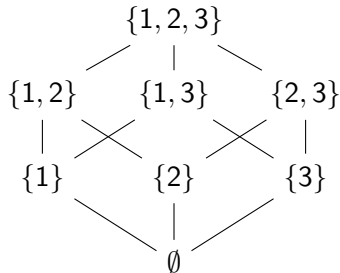
Hasse Diagram of $(P(A), \subseteq)$

Let

$$A = \{1, 2, 3\}.$$

Then

$$P(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}.$$



Inclusion–Exclusion Principle

Definition

The Principle of Inclusion–Exclusion is used to find the number of elements in the union of overlapping finite sets.

For two finite sets A and B ,

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

For three finite sets,

$$\begin{aligned} |A \cup B \cup C| &= |A| + |B| + |C| \\ &\quad - |A \cap B| - |A \cap C| - |B \cap C| \\ &\quad + |A \cap B \cap C|. \end{aligned}$$

The signs alternate between addition and subtraction.

General Principle of Inclusion–Exclusion

Theorem

Let $A_1, A_2, \dots, A_n$ be finite sets. Then

$$\begin{aligned} |A_1 \cup A_2 \cup \dots \cup A_n| = & \sum |A_i| - \sum |A_i \cap A_j| \\ & + \sum |A_i \cap A_j \cap A_k| - \dots \\ & + (-1)^{n+1} |A_1 \cap A_2 \cap \dots \cap A_n|. \end{aligned}$$

The pattern is

+single sets – pairwise intersections + three-way intersections – ...

Proof of the General Principle

Suppose an element x belongs to exactly r of the given sets.

In the first summation, x is counted once for each set containing it.

Hence, it is counted

$$\binom{r}{1}$$

times.

In the second summation, it is subtracted once for every pair of sets containing it.

Hence, it is subtracted

$$\binom{r}{2}$$

times.

Proof: Continuing

Similarly,

$$+ \binom{r}{3}$$

is added from the three-set intersections,

$$- \binom{r}{4}$$

is subtracted from the four-set intersections, and so on.

Therefore, the total number of times x is counted is

$$\binom{r}{1} - \binom{r}{2} + \binom{r}{3} - \cdots + (-1)^{r+1} \binom{r}{r}.$$

Binomial Identity

Using the binomial theorem,

$$(1 - 1)^r = 0.$$

Therefore,

$$\binom{r}{0} - \binom{r}{1} + \binom{r}{2} - \cdots + (-1)^r \binom{r}{r} = 0.$$

Since

$$\binom{r}{0} = 1,$$

we obtain

$$\boxed{\binom{r}{1} - \binom{r}{2} + \binom{r}{3} - \cdots + (-1)^{r+1} \binom{r}{r} = 1.}$$

Proof: Conclusion

Since every element belonging to the union is counted exactly once,

$$|A_1 \cup A_2 \cup \cdots \cup A_n|$$

is equal to

$$\begin{aligned} \sum |A_i| - \sum |A_i \cap A_j| + \sum |A_i \cap A_j \cap A_k| - \cdots \\ + (-1)^{n+1} |A_1 \cap A_2 \cap \cdots \cap A_n|. \end{aligned}$$

Hence,

The General Principle of Inclusion–Exclusion is proved.



AI/ML Example of Inclusion–Exclusion

Suppose in an AI course:

$$A = \{\text{students studying CNN}\},$$

$$B = \{\text{students studying RNN}\},$$

$$C = \{\text{students studying Transformers}\}.$$

Suppose

$$|A| = 40, \quad |B| = 30, \quad |C| = 25,$$

$$|A \cap B| = 10, \quad |A \cap C| = 8, \quad |B \cap C| = 7,$$

and

AI/ML Example: Solution

Using the three-set Inclusion–Exclusion formula,

$$\begin{aligned}|A \cup B \cup C| &= |A| + |B| + |C| \\ &\quad - |A \cap B| - |A \cap C| - |B \cap C| \\ &\quad + |A \cap B \cap C|.\end{aligned}$$

Substituting,

$$|A \cup B \cup C| = 40 + 30 + 25 - 10 - 8 - 7 + 3.$$

Therefore,

$$\boxed{|A \cup B \cup C| = 73}.$$

Thus, 73 students study at least one of CNN, RNN or Transformers.

Important Concepts: Summary

- A set is a well-defined collection of distinct objects.
- Set-builder form describes a set using a condition.
- Roster form lists the elements explicitly.
- Set operations include union, intersection, difference, symmetric difference, complement and Cartesian product.
- A relation is a subset of a Cartesian product.
- An equivalence relation is reflexive, symmetric and transitive.
- A partial order is reflexive, antisymmetric and transitive.
- A Hasse diagram represents a finite poset.
- A chain contains comparable elements.
- An antichain contains mutually incomparable elements.
- Inclusion–Exclusion counts elements in overlapping sets.

Thank You!

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